

Damage

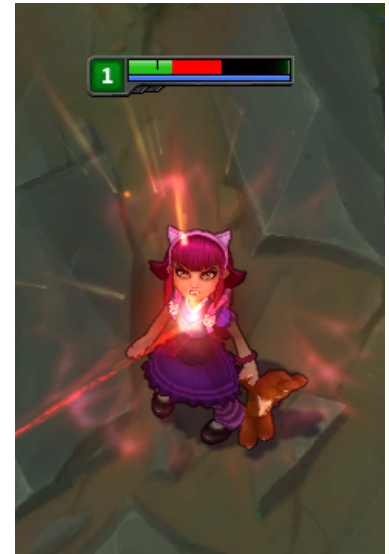


This article or section is substantially or structurally incomplete. You can help the League of Legends Wiki by [expanding](#) it. For more projects, see the [Community To Do List](#).

Additional information: This article and the **Damage types** section in particular contain confusing or outdated information that is presently subject to a complete overhaul.

Note: Not to be confused with [attack damage](#), a champion statistic, nor [health costs](#).

Damage is the deduction of a unit's current health as a result of an offensive interaction referred to as a **damage instance** or **damage event**.



Damage is displayed as a red section on a champion's health bar.

Contents

[Damage type classifications](#)

[Damage types](#)

[Damage tags](#)

[Damage properties](#)

[Damage values](#)

[Base values and character statistics ratios](#)

[Critical strike](#)

[Successful ability casts](#)

[Capped damage](#)

[Damage value modifiers](#)

[Invulnerability](#)

[Calculating mitigation](#)

[Calculating applied damage](#)

[Engine level](#)

[Script level](#)

[Subtypes](#)

[Resistance subtypes](#)

[Calculating Damage](#)

[Capped Damage](#)

Misconceptions

See also

Damage type classifications

Damage events in League of Legends can be assigned different **types**, **tags** and **properties** to manage interactions, such as mitigation by the respective type of resistances or additional effects to be triggered by the damage event based on tags.

Damage types

Any instance of damage can have only one of the following types:

Physical damage

Mitigated by  armor. Almost all units' basic attacks deal exclusively **physical damage**.

Magic damage

Mitigated by  magic resistance. The powerful ability casts of  Mages deal **magic damage**.

True damage

Not reduced by resistances nor almost all damage reduction effects. Reserved for powerful effects or those that for gameplay or technical reasons are not supposed to interact with resistances or reduction effects. True damage still respects  shields and  invulnerability.

Damage tags

Any instance of damage can have none or multiple of the following tags:

BasicAttack

A tag found on the primary damage instance of basic attacks including empowered forms of basic attacks (e.g.  Blitzcrank's  Power Fist,  Darius'  Crippling Strike), attack-like spells (e.g.  Ezreal's  Mystic Shot,  Fiora's  Lunge) and automatic effects behaving like attacks (e.g.  Akshan's  Heroic Swing,  Bel'Veth's  Royal Maelstrom,  Urgot's  Purge,  Runaan's Hurricane). Damage with this tag will interact with  Warden's Mail.

ActiveSpell

A tag found on damage instances of most abilities and some systemic effects that makes them trigger **spell effects**, including most empowered basic attacks. Often associated with the user spending mana and/or a cooldown to deal the damage.

AOE

A tag that lets the game recognize an instance of damage as coming from an *area of effect* source. Certain  pets are more resilient against damage with this tag in order to make them less susceptible against champions with strong area-of-effect abilities. Often accompanies the **ActiveSpell** tag.

Periodic

A tag that lets the game recognize an instance of damage as a repetitive kind. Found on Damage debuffs (e.g.  Malzahar's Malefic Visions,  Singed's Poison Trail,  Liandry's Torment), most non-targeted spells that deal damage repeatedly (e.g.  Lucian's The Culling,  Morgana's Tormented Soil). Damage with the **periodic** tag does not interrupt  sleep.

Item

A tag that lets the game recognize an instance of damage as coming from a non-champion source, such as an item or summoner spell. This allows effects that are supposed to only trigger on champion abilities (e.g.  Tiamat's Cleave,  Liandry's Torment's Torment) to ignore damage instances with this tag.

Proc

A tag that lets the game recognise an instance of damage as being applied as a secondary effect. This is commonly used for on-hit effects or other instances of damage that shouldn't interact with anything else. This tag is normally reserved for automatic applications of **on-hit effects** (e.g.  Jax's Grandmaster-At-Arms,  Kog'Maw's Bio-Arcane Barrage,  Blade of the Ruined King).

Pet

NonRedirectable

Indirect

Damage properties

Any instance of damage can have none or multiple of the following properties:

ApplyLifesteal

A tag that forces a damage instance to apply life steal. Unless otherwise specified it applies at 100% effectiveness, with no penalty for AoE damage. Found on certain auto-attack-like abilities (e.g.  Samira's Flair,  Camille's Precision Protocol,  Hecarim's Devastating Charge).

EnableCallForHelp

RespectImmunity

RespectDodge

A tag that forces a damage instance to respect dodges (e.g.  Nilah's Jubilant Veil,  Jax's Counter Strike). This is the default for basic

damage, as well as some abilities.

TriggerOnHitEvents

TriggerDamageEvents

ApplyOmnivamp

Nameless property

Damage values

Base values and character statistics ratios

Critical strike

A damage instance can **critically strike** when it is implemented as such on an active attack or ability. Damage that critically strikes has been modified by a percentage above 100%, thus deals increased damage. Critical strikes interact exclusively with  [critical strike chance](#),  [critical strike damage](#), and  Randuin's Omen's Resilience. The highest critical strike damage instance is also tracked in the post-game statistics.



Physical damage
critical strike



Magic damage
critical strike



True damage
critical strike

A [basic attack](#) that critically strikes is modified by the attacker's  [critical strike damage](#), which is by default 175%. Most on-hit effects will not be affected, unless specified (e.g.  Riven's  Runic Blade).

 [Critical strike chance](#) above 100% does not affect critical strikes.

Average damage calculations

Assuming:

All amounts are in decimal form.

DamageBase = the normal damage amount (before any modifiers)

CritChance = critical strike chance

CritMod = critical strike modifier/ratio

Then the average damage per attack when continuously attacking can be calculated with the formula:

$$\text{DamageBase} + \text{CritChance} \times (\text{CritMod} - 1) \times \text{DamageBase}$$

For example:

100 DamageBase, 40% CritChance, 175% CritMod

$$100 + 0.4 \times (1.75 - 1) \times 100 = 130 \text{ average damage per attack}$$

100 DamageBase, 100% CritChance, 175% CritMod

$$100 + 1.0 \times (1.75 - 1) \times 100 = 175 \text{ average damage per attack}$$

100 DamageBase, 80% CritChance, 215% CritMod

$$100 + 0.8 \times (2.15 - 1) \times 100 = 192 \text{ average damage per attack}$$

100 DamageBase, 100% CritChance, 120% CritMod

$$100 + 1.0 \times (1.20 - 1) \times 100 = 120 \text{ average damage per attack}$$

Successful ability casts

Certain abilities have unique impact text that triggers when they deal increased damage due to the player successfully passing the ability's condition (e.g. to denote  Karthus'  Lay Waste hitting a single target). On the LoL Wiki, these are referred to as  cosmetic critical strikes (not to be confused with actual  critical strikes). A list of all abilities that use this impact effect can be found below.



Physical damage
"successfully
enhanced"



Magic damage
"successfully
enhanced"



True damage
"successfully
enhanced"

 Aatrox's  The Darkin Blade against enemies hit by each cast's **Sweetspot**.

 Anivia's  Frostbite against enemies recently hit by  Flash Frost or a fully formed  Glacial Storm

 Aurelion Sol's  Breath of Light against bursted enemies and  The Skies Descend against enemies hit

 Bel'Veth's  Endless Banquet against champions with her attacks

 Briar's  Chilling Scream against enemies that hit a wall

 Caitlyn's  Piltover Peacemaker against enemies hit if they were revealed by  Yordle Snap Trap

 Evelynn's  Empowered Whiplash against the target and  Last Caress against enemies under the health threshold

 Fiddlesticks'  Terrify against the target if they were recently feared by the ability

-  Gnar's  GNAR! against enemies that collide with terrain
-  Gragas'  Barrel Roll against enemies hit after the barrel has fully fermented
-  Gwen's  Snip Snip! against enemies hit in the center of the cone
-  Hwei's  Severing Bolt against immobilized or isolated targets
-  Karthus'  Lay Waste against an isolated enemy
-  Kog'Maw's  Living Artillery against enemies under the health threshold
-  LeBlanc's  Sigil of Malice,  Distortion, and  Ethereal Chains against enemies marked by  Sigil of Malice
-  Lillia's  Watch Out! Eep! against enemy champions and monsters hit in the center of the area
-  Lulu's  Glitterlance against enemies hit by both bolts
-  Malphite's  Thunderclap against the target hit by the first attack
-  Mordekaiser's  Obliterate against a single target
-  Naafiri's  Darkin Daggers against the target if they are already bleeding
-  Nidalee's  Takedown against the target if they were  Hunted
-  Pantheon's  Comet Spear against enemies under the health threshold
-  Pyke's  Death from Below against executed enemy champions
-  Qiyana's  Elemental Wrath (**TERRAIN**) against enemies under the health threshold
-  Rek'Sai's  Furious Bite at maximum Fury
-  Ryze's  Overload against enemies marked by  Spell Flux
-  Samira's  Taunt against the enemy hit
-  Shaco's  Two-Shiv Poison against enemies under the health threshold
-  Seraphine's  High Note against enemies that take increased damage
-  Sett's  Haymaker and  The Show Stopper both against enemies hit in the center of the area
-  Smolder's  MMOOOMMMM! against enemies in the center
-  Taliyah's  Threaded Volley when casting on **Worked Ground**
-  Udyr's  Awakened  Wilding Claw against both the target and enemies hit by the lightning, and  Awakened  Wingborne Storm bonus damage.
-  Urgot's  Fear Beyond Death against the executed target
-  Varus'  Piercing Arrow against enemies hit if the ability was fully charged (includes  Blighted Quiver's bonus damage)
-  Vayne's  Condemn against the target if they collide with terrain
-  Vex's  Doom 'n Gloom against targets marked with *Gloom*
-  Vladimir's  Transfusion against the target if the ability was empowered by  Crimson Rush
-  Volibear's  Frenzied Maul against the target if they are *Wounded* and
-  Stormbringer against enemies hit in the center of the area
-  Yorick's  Mist Walkers upon attacking enemies marked by  Mourning Mist

-  Zeri's  Basic Attack against executed targets
-  Ziggs'  Satchel Charge against turrets below the health threshold
-  Rift Herald's charge against turrets
-  Serpent's Fang's Shield Reaver against  shields

Capped damage

Damage value modifiers

See also: Damage modifier

Invulnerability

See also: Invulnerability

Calculating mitigation

See also: Armor and Magic Resistance

Calculating applied damage

See also: Shields

The information below describes an outdated model of damage classification, but will continue to exist until the section and cross-wiki references are fully overhauled.

There are two systems for classifying damages: an engine-level system and a script-level system. As the name suggests, the engine-level system is integrated into the game's engine; while the script-level system is a relatively modern addition that allows Riot more flexibility.

Damage classifications modify what effects are triggered when damage is dealt or received - e.g. on-hit effects (such as  Wit's End) are triggered whenever your champion deals a source of  basic damage.

Engine level

Wiki Name	API name	Description	Example(s)
 <u>Raw damage</u> ^e	raw	Damage that ignores shields. It is hard-coded to ignore  <u>resurrection</u> , but specific instances are	 <u>Fear Beyond Death</u> ,  <u>Death from Below</u> and  <u>Nexus Obelisk</u> (to non-champions)

		then special-cased to be un-ignored.	
N/A	internalraw	As above, but also ignores  invulnerability,  Resurrection and  Bailout are ignored.	 Nexus Obelisk (to champions),  Rift Herald's self-damage,  Cell Division's redirected damage, and  Friend of the Forest's grove freeing to monsters
 Default damage ^e	periodic ^[s 1]	Ability-based procs with no interactions. Effects without a proper damage type default to this category, including most splash damage.	 Umbral Blades,  Ricochet,  Liandry's Torment,  Tiamat,  Ignite
 Proc damage ^e	proc ^[s 2]	Attack-based procs with no interactions. Proc effects are triggered whenever you deal <i>basic damage</i> .	 Wit's End
 Reactive damage ^e	reactive	Damage type dealt by Thorns. Similar reflection effects (e.g. Annie and Rammus) are classified as area damage.	Only  Thornmail and  Bramble Vest
 Basic damage ^e	attack ^[s 3]	Damage dealt by any unit(s), which includes turrets, neutral monsters, and minions <u>basic attack</u> and some champion abilities.	 Mystic Shot,  Lunge,  Steel Tempest
 Spell damage ^e	spell ^[s 4]	Damage used for single target abilities and	 Disintegrate,  Ace in the

		some basic attacks.	Hole,  Noxian Diplomacy,  Pick a Card
 <u>Area damage</u> ^e	spellaoe	Damage dealt by most abilities <i>capable</i> of damaging multiple units.	 Incinerate,  Rupture,  Eye of Destruction
 <u>Persistent damage</u> ^e	spellpersist	Damage dealt through DoTs	 Blaze,  Malefic Visions
 <u>Pet damage</u> ^e	pet	Damage dealt by most pet basic attacks. Other pet damage is classified as Spell/ SpellPersist.	 Tibbers,  Daisy,  Eulogy of the Isles

1. The League of Legends Wiki decided that the word "periodic" would be too easily misunderstood due to the implication of 'bleed' or 'over-time' behaviour.
2. The term 'proc' refers to a mechanic that has a chance of triggering, which is common in games such as World of Warcraft (e.g., "Your attacks have an x % chance to cause [...] effects). League of Legends originally featured 'true' proc effects (e.g., "your attacks have a chance to slow.") However, the design philosophy of Riot Games' over-time grew more towards preferring 'on-hit' or "every nth attack" to purely random mechanics. Despite this: the engine-level label persisted. The term 'proc' now describes all additional effect triggers even if they are not guaranteed, 'proc' is used as a verb when triggering such an effect.
3. The League of Legends Wiki elected to use the 'basic' portion of basic attack to refer to this damage type to avoid confusion between 'attack damage' the damage type and attack damage the statistic.
4. Some abilities such as Karthus' Lay Waste are capable of dealing either spell damage or area damage depending on the number of targets hit.

Before our editors determined the above damage types from the game files, the wiki's damage type documentation was based on observation. The wiki previously grouped *spell*, *spellaoe* and *spellpersist* as a single 'ability damage' type, with 'single target,' 'aoe' and 'dot' being noted as subtypes. The wiki also used *pure damage* as a conjectural placeholder for the now-termed *raw damage*. While the described functionality was usually correct, the previous documentation included many 'special cases' for abilities that did not behave consistently with the observed rules.

Despite having distinct use-cases,  proc damage and  default damage are almost indistinguishable in terms of their gameplay functionality. Differentiating the two damage types without accessing the game files is extremely difficult for some effects. One of the only known ways to distinguish the two types involves  Teemo: Instances of *default damage* do not disable  Move Quick's passive movement speed, while instances of *proc damage* will. This distinction does not apply to other "out of combat" effects.

Script level

The following are the currently known script-tags:

AOE
Periodic
Indirect
BasicAttack
ActiveSpell
Proc
Pet
NonRedirectable
Item

For the most part, the two systems correlate and so the League of Legends Wiki has elected to not to document the script-level system separately across all articles. Instances where script-level tags create deviations from engine-level rules will be noted on a case-by-case basis in the ability-details. A notable but consistent deviation is  [persistent area damage](#), as documented below:

Wiki Name	API name	Description	Example(s)
 Persistent area damage ^e	N/A	Abilities classified as area damage on an engine-level, but periodic on a script level. This means that engine-level effects (such as spell vamp) trigger the reduced healing, while script-level effects (such as  Arcane Comet) trigger the damage-over-time cooldown reduction.	 Darkness Rise ,  Tormented Shadow ,  The Equalizer

Subtypes

Resistance subtypes

See also: [Armor and Magic Resistance](#)

Resistance subtypes are a secondary typing system that exists to promote [champion classes](#) by diversifying the effectiveness of items and champion match-ups. There is a small selection of effects that are triggered by specific subtypes rather than the above archetypes - e.g.  [Morellonomicon](#) will trigger on anything with the [magic damage](#)

subtype, including abilities, basic attacks and on-hit effects (i.e.  Wit's End will trigger Morellonomicon on a champion with no other source of magic damage). These are the damage types visible to players in-game.

Name	Description	Example
 <u>Physical damage</u>	A damage type thematically associated with making physical contact, such as fists, swords and the physical projectiles of bows and guns.  Armor reduces incoming physical damage.	Most basic attacks; most AD-scaling abilities; some item effects.
 <u>Magic damage</u>	A damage type thematically associated with non-physical, such as magic or the supernatural.  Magic resistance reduces incoming magic damage.	Most AP-scaling abilities; some item effects.
 <u>True damage</u>	A damage type thematically associated with bypassing the target's defenses. Not to be confused with  <u>raw damage</u> .	A select few sources, such as  <u>Smite</u> ,  <u>Ignite</u> ,  <u>Silver Bolts</u> and  <u>Reckless Swing</u> .

Calculating Damage

Also see: Damage modifier.

The following effects influence damage as non-zero values, in no particular order of calculation. They may originate from the attacker or the target.

- Percentage damage amplification
- Percentage damage reduction
- Flat damage reduction
- Resistances, meaning  armor and  magic resistance, against **physical damage** and **magic damage**, respectively
- Resistances reduction
- Resistances penetration, meaning  armor penetration and  magic penetration

Resistances effectively reduce the respective damage subtype (physical or magic) by a **percentage**. However, "**percent damage reduction**" is a different, independent effect: abilities such as  Alistar's Unbreakable Will and item effects such as  Plated Steelcaps' *Plating* may grant it as a beneficial effect while effects such as  Exhaust may apply it as a debuff. Percent damage modifiers  stack multiplicatively with the damage reduction achieved through resistances.

After the physical and magic damage is affected by the various reductions, it is applied to the target in  health amount. This amount received by the target is called *post-mitigation damage*.

Capped Damage

Some abilities, scaling with target's health, have a capped amount of damage they can deal to a target. This is made to avoid dealing massive damage to targets with a lot of health, most often against only  minions and/or  monsters.

If the ability deals enough damage (before damage reductions) to reach the cap, the ability's damage is reduced to the cap and damage reductions are then applied. This means that against a target that has negative armor or magic resistance the ability's post-mitigation damage can be higher than the cap.

Misconceptions

The resistance types—physical, magic and true—are often regarded by players to be the archetypal damage types, which leads to a lot of misconceptions.

For example, some players believe that magic damage, ability damage and AP are all interchangeable terms, with even prominent streamers referring to abilities as dealing "AP damage" and "AD damage". This is incorrect. It is possible for basic attacks to deal magic damage (e.g.  Thresh with a point in  Flay); it is possible for abilities to deal physical damage (e.g.  Riven); it is possible for AD-scaling abilities to deal magic damage (e.g.  Shyvana); and it is possible for non-magic damage to apply spell effects (e.g.  Luden's Companion would trigger on Riven if she chose to buy it, although she's discouraged due to her having no AP scalings).

Similarly, many players believe that life steal and spell vamp are specific to physical and magic damage, respectively. Life steal will apply to anything classified as basic damage (including the magic damage from  Takedown and  Infinite Duress and true damage from  Hextech Munitions), and spell vamp will apply to anything classified as ability damage (including the physical damage from  Comet Spear and true damage from  Reckless Swing).

See also

Damage modifier

Magic damage

Physical damage

True damage

Gameplay Elements

[show]

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